

Question Number	Answer	Additional guidance	Mark
5(a)(i)	An answer that makes reference to the following points: • {clearing of land with / cutting down of} Boabab trees {to grow rice / to grow crops / for animal grazing / for agriculture} (1) • low pollination rate due to {fewer fruit bats / larger distances between trees} (1)	ignore competition with rice plants ignore Boabab trees are cut down unqualified ignore deforestation unqualified ignore disease ACCEPT (remaining) trees are {isolated / far apart} so reducing chances of {pollination / sexual reproduction} ACCEPT Boabab trees cannot reproduce asexually	(2)

Question Number	Answer	Additional guidance	Mark
5(a)(ii)	An answer that makes reference to the following points: • Madagascan people will (be taught to) not cut down the Baobab trees (1) • Madagascan people will (be taught to) promote the planting of {seeds / saplings} to increase the population (1)	ACCEPT reduce {number of trees cut down / deforestation / animals eating young plants / hunting of bats} ignore prevent habitat destruction ACCEPT protection of bats ACCEPT more trees will be {planted / grown} / reforestation of baobab / artificial pollination (of baobab) ignore captive breeding programmes / seed banks	(2)

Question Number	Answer	Additional guidance	Mark
5(b)(i)	An explanation that makes reference to the following points: - calcium pectate (1) {hold cell walls together / increase cell wall stability / form middle lamella / hold cellulose microfibrils together} (1)	ACCEPT pectin / pectate ACCEPT to increase {cell wall / tensile} strength ACCEPT increased support ignore make cell walls	(2)

Question Number	Answer	Mark
5(b)(ii)	The only correct answer is A phloem <i>B is not correct because sclerenchyma do not transport water down the stem of a plant</i> <i>C is not correct because vacuoles do not transport water down the stem of a plant</i> <i>D is not correct because xylem do not transport water down the stem of a plant</i>	(1)

Question Number	Answer	Additional guidance	Mark
5(c)	An answer that makes reference to the following points: - there were more pods per soybean plant when magnesium ions are present (1) - there were more pods per soybean plant in 2020 than 2019 (1)	ACCEPT converse for no added magnesium ACCEPT there are more pods as time passes ACCEPT converse	(2)

Question Number	Answer	Additional guidance	Mark
5(d)	An answer that includes four of the following points: • presence of magnesium ions increased the mass of both (maize grains and soybeans) (1) • mass of maize is greater than the mass of soybean / converse (1) • the percentage increase in mass of soybeans was greater than maize (1) • significant difference between maize and soybean due to no overlapping of {error / SD} bars (1) • relevant comment about methodology (1)	ACCEPT positive correlation ACCEPT difference is significant for soybean (with and without magnesium) due to no overlap of {error/ SD} bars / converse for maize ignore without clear statement which plant(s) they are referring to e.g. was same concentration of Mg^{2+} added	(4)